

Can you give me list of all dsa topic

Since you're learning C# and aiming to become a .NET Full Stack Developer, here's a complete Data Structures and Algorithms (DSA) roadmap from beginner to advanced.

1. Complexity Analysis

- Time Complexity
- Space Complexity
- Big O Notation
- Big Omega (Ω)
- Big Theta (Θ)
- Best, Average, Worst Case

2. Arrays

- 1D Array
- 2D Array
- Jagged Array
- Dynamic Array
- Traversal
- Insertion
- Deletion
- Searching
- Rotation
- Prefix Sum
- Sliding Window
- Kadane's Algorithm

3. Strings

- String Manipulation
- Pattern Matching
- Palindrome
- Anagrams
- String Builder
- Rabin-Karp
- KMP Algorithm
- Z Algorithm

4. Recursion

- Basic Recursion
- Tail Recursion
- Backtracking
- Recursive Tree Analysis

5. Linked Lists

- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Reverse Linked List
- Detect Loop
- Floyd Cycle Detection
- Merge Lists

6. Stack

- Stack Operations
- Stack Using Array
- Stack Using Linked List
- Expression Evaluation
- Balanced Parentheses
- Monotonic Stack

7. Queue

- Simple Queue
- Circular Queue
- Priority Queue
- Deque
- Queue Using Stacks

8. Hashing

- Hash Table
- HashSet
- Dictionary
- Collision Handling
- Frequency Counting

9. Searching Algorithms

- Linear Search
- Binary Search

- Ternary Search
- Lower Bound
- Upper Bound

10. Sorting Algorithms

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Heap Sort
- Counting Sort
- Radix Sort
- Bucket Sort

11. Trees

- Tree Basics
- Binary Tree
- Binary Search Tree (BST)
- AVL Tree
- Red-Black Tree
- Segment Tree
- Fenwick Tree (BIT)
- Trie

12. Tree Traversal

- Preorder
- Inorder
- Postorder
- Level Order
- Zig-Zag Traversal
- Boundary Traversal

13. Heap

- Min Heap
- Max Heap
- Heapify
- Priority Queue
- K Largest/Smallest Elements

14. Graphs

- Graph Representation
- BFS
- DFS
- Connected Components
- Cycle Detection
- Topological Sort
- Bipartite Graph

15. Shortest Path Algorithms

- Dijkstra
- Bellman-Ford
- Floyd-Warshall
- A* Algorithm

16. Minimum Spanning Tree

- Prim's Algorithm
- Kruskal's Algorithm
- Disjoint Set Union (DSU)

17. Greedy Algorithms

- Activity Selection
- Huffman Coding
- Fractional Knapsack
- Job Scheduling

18. Dynamic Programming (DP)

- Memoization
- Tabulation
- 0/1 Knapsack
- Longest Common Subsequence
- Longest Increasing Subsequence
- Matrix Chain Multiplication
- DP on Trees
- DP on Graphs

19. Backtracking

- N Queens
- Sudoku Solver

- Rat in Maze
- Permutations
- Combinations

20. Bit Manipulation

- Bitwise Operators
- XOR Tricks
- Set/Clear Bit
- Power of Two
- Bitmasking

21. Advanced Data Structures

- Trie
- Segment Tree
- Fenwick Tree
- Sparse Table
- B Tree
- B+ Tree
- Skip List

22. Advanced Algorithms

- Divide and Conquer
- Meet in the Middle
- Mo's Algorithm
- KMP
- Rabin-Karp
- Z Algorithm
- Manacher's Algorithm

23. Mathematical Algorithms

- Prime Numbers
- Sieve of Eratosthenes
- GCD
- LCM
- Modular Arithmetic
- Fast Exponentiation
- Combinatorics

24. System Design Related DSA

- LRU Cache
 - LFU Cache
 - Consistent Hashing
 - Bloom Filter
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Recommended Order for You

1. Complexity Analysis
2. Arrays
3. Strings
4. Recursion
5. Linked Lists
6. Stack
7. Queue
8. Hashing
9. Searching
10. Sorting
11. Trees
12. Heap
13. Graphs
14. Greedy
15. Backtracking
16. Dynamic Programming
17. Bit Manipulation
18. Advanced Topics

For a .NET developer preparing for interviews, focus heavily on:

- Arrays
- Strings
- Linked Lists
- Stack & Queue
- Hashing
- Trees
- Graphs
- Dynamic Programming
- Sorting & Searching

These topics cover about 90% of common coding interview questions.

